

I have seen many posts about how much energy was used for each query on AI, but nowadays, google have that gemini AI auto-preview for everything you search. Does this mean google something now take more energy than ever?

Posted by u/GrayNish • 0 points • 7 comments

And googling is no less wasteful than using chatgpt then?

Comments

u/eggs-benedryl • 2 points • 2025-05-15 18:23 UTC

Yes and we don't know which is more wasteful until we look at the actual numbers. Which I don't have. I have read swearing gets the ai overview to stop.

That being said, much of Ai research involves trying to get the best performance from smaller models. To optimize everything, how much time and therefore energy is used to train, how hard they are to run, what tools can be used to enhance smaller models rather than just making bigger models that are less efficient.

All of these companies (except CloseAI heh), work on free open source models small enough to run on your own PC or even phone. Using no more energy than other short but demanding tasks.

If you hook that up to wikipedia or a reliable source of data then you dont need a big model to give good answers if you took a small model good at REASONING and gave it more info then it can likely perform the same tasks a massive model could.

For perspective, google/openai/meta all have models 500B plus, that's billions of parameters. These same companies are working to scale things down if it helps and still gets similar/the same performance. They each have released models as small as 0.5B that are still technically capable for certain things. If you can run a teeeeeny tiny model and artificially make it very smart then your energy use can plummet.

u/Lollygagg • 1 points • 2025-08-13 20:09 UTC

"swearing gets the ai overview to stop."

thanks for the tip to remove the Ai Overview

I had the opportunity to ask this broader energy question of the lead at either Google Gemini or Google Labs during a donor-related panel conference, shortly after Ai lab answers were rolled out to auto-answer search queries. I asked what we can currently assume to be the power consumption difference between the 3 search modes. If not the difference, than at least the ranking of the 3 options so that we can make educated choices.

He skirted the question, first pausing... and the host added subterfuge by chiming in with something irrelevant for him (even though it was out of her purview), just to make him feel comfortable. Than he replied with an unrelated, but attractively similar acting answer and all sorts of emotionally sensitive language to act like he was reaching towards me with an answer hug...along with the customary-- "we are all working everyday to maximize our efficiency" It was such an obvious swerve and disarm moment that it gives me little confidence about it all. A retired, upper level politician walked up to me afterwards and reiterated that he and his wife saw all of that and were glad I asked anyways.

u/TheApiary • 2 points • 2025-05-15 18:14 UTC

Yeah. But the good news is, that amount of energy is really small. People online have been making it sound big by saying "It's like 10 google searches!" which is true, but 10 google searches is about as much energy as running a microwave for 10 seconds. So if you aren't fussed about the difference between microwaving something for 50 seconds versus a minute, because you know that's a tiny amount of energy that doesn't functionally make any difference to anything, then you don't need to worry about this either.

An analogy I like is it's like saying "Red bell peppers are so high in calories, they have almost twice as many calories as green bell peppers!" It's true that they have almost twice as many calories as green bell peppers, but the green ones only have like 20 calories per pepper and the red have like 40, which is still really small. So if someone took away from that "I'm trying to lose weight so I should avoid red bell peppers" that probably wouldn't work well, because red bell peppers almost definitely aren't a major source of calories in their diet.

u/the_calibre_cat • 1 points • 2025-07-30 17:39 UTC

> People online have been making it sound big by saying "It's like 10 google searches!" which is true, but 10 google searches is about as much energy as running a microwave for 10 seconds.

bruv no

Microwaves use anywhere between 800-1800 watts (depending on model) - whereas a standard Google search [is reportedly around 0.0003 kWh] (<https://googleblog.blogspot.com/2009/01/powering-google-search.html>). Microwaves vs. a standard Google search aren't remotely in the same ballpark, and that 0.0003 kWh comes from 2009 - performance-per-watt has *dramatically* improved since then, and we're talking about something that isn't 50 seconds vs 1 minute, we're talking about something that's more like 50 seconds vs nearly one hour (~3,333 seconds, by my math), or nearing two orders of magnitude more.

As for AI queries, they're a little more subdued, but most estimates put them at AT LEAST 10x the energy consumption of a standard Google search, so one order of magnitude (50 seconds vs 500 seconds) - but [MIT seems to think that the energy cost is much, much higher] (<https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-footprint-big-tech/>), and [we have some data straight from the horse's mouth] (<https://huggingface.co/spaces/AIEnergyScore/Leaderboard>), at least as far as semi-modern AI

models are concerned. A lot of the more modern AI models are not featured in that dataset, because their developers have opted not to share them. Which is... concerning, to say the least.

So like, I hear your response here, but your numbers just aren't even in the same ballpark as to the real-world differences in energy intensity between a standard web search vs. an AI query.

u/TheApiary • 1 points • 2025-07-30 18:38 UTC

Let's call the microwave 1000 Watts just for simplicity's sake. In that case, if you ran it for 1 hour, it would use 1000 Wh. If you ran it for 10 seconds it would use about 2.78 Wh.

The number I saw in the thing you linked estimated 1 ChatGPT query at about 0.3 Wh.

Image generation models do use more than that most likely

u/the_calibre_cat • 1 points • 2025-07-30 19:42 UTC

> The number I saw in the thing you linked estimated 1 ChatGPT query at about 0.3 Wh.

That's pretty much exactly on-par with a Google search... from 2009.

> Image generation models do use more than that most likely

Per my second link, that does indeed appear to be the case. Not terribly surprising.

u/Kakamile • 1 points • 2025-05-15 18:23 UTC

There are approximately 16.4 billion searches on Google every day

u/OfficialHashPanda • 1 points • 2025-05-15 19:55 UTC

That's another way of showing how tiny the energy usage of google searches is.

u/TheApiary • 1 points • 2025-05-15 19:37 UTC

That's true, but I think OP's question was about whether they should feel like they are wasting energy when they google something. And I'm saying that if you wouldn't have felt bad about doing 10 google searches last year, and you don't feel bad about microwaving something for an extra 10 seconds, then it would be kind of weird to feel bad about this.

Also, even the total amount is much less than it might sound in energy use terms. If my math is right, that's about 4.8 million kwh per day on Google searches with Gemini. Global energy consumption in 2023 was around 500,000 GWh per day.

Which would mean that Google with Gemini takes up about 0.000001% of the global energy use. Which means that even if Google completely stopped doing searches, there wouldn't be a noticeable drop in how much energy the world uses, because the fraction that Google search uses is such a tiny drop in the bucket.

It would be more if you included non-generative AI, like all the AIs that recommend ads and videos to us, but that's not what this question was about.

u/GrayNish • 1 points • 2025-05-15 20:59 UTC

Now i wonder, if you dont mind, does the statement of "it equal to 10 google searches" refer to ai before or after google implemented their own gemini.

If it's the former, does that still hold true?

If it's the latter, is gemini that much more efficient compared to other generative ai?

u/TheApiary • 1 points • 2025-05-15 21:50 UTC

The thing where it equals 10 google searches is pre gemini, just normal old fashioned google searches. My math in that comment was accounting for that though, I treated a new google search as the same as a chatGPT query or 10 old google searches.